

Pathway Database



$$\Sigma_{0j}^{(A)} = A \odot \Sigma_{0j}$$

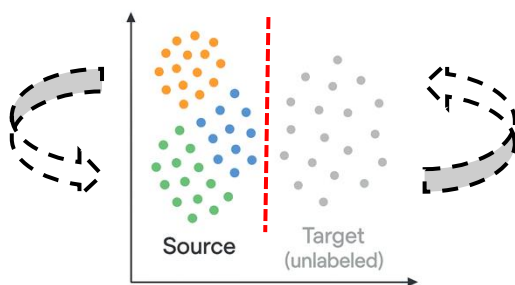
Source Dataset*



Large-scale
Biobanks/Atlases,
or an Auxiliary
Upstream Omic
Layer

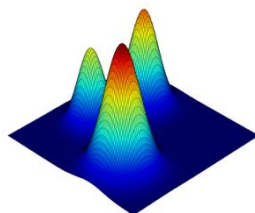


Robust
Cluster
Structure



*Domain Heterogeneity
for Datasets with
Different Individuals*

Constructing Prior



$$p(z) = \sum_{j=1}^J \theta_{0j} N_P(\mu_{0j}, \Sigma_{0j}^{(A)})$$

Target Dataset*

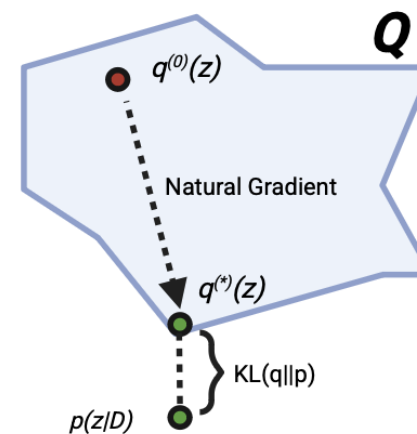


Small Cohort
Studies, or a
Downstream
Omics Layer



To Be
Clustered

PRAXIS-BGM



**Natural-gradient
Variation Inference**

Posterior Clusters



*Data points may be subject-level or cell-level (scRNA-seq).